Stokes equation (simplified form)

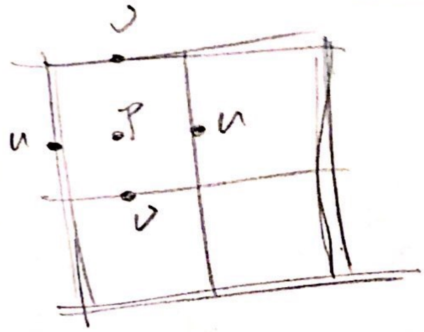
$$\begin{cases} -\Delta \vec{u} + \nabla p = \vec{f} \\ -\nabla \cdot \vec{u} = g \end{cases} \quad \text{in } \Omega$$

$\operatorname{div} \vec{u} = \frac{\partial \vec{u}}{\partial x} + \frac{\partial \vec{u}}{\partial y}$ in 2-dimension Ω.

$\vec{u} = (u, v)$, $\vec{f} = (f_1, f_2)$

$$\Rightarrow \begin{cases} -\Delta u + p_x = f_1 \\ -\Delta v + p_y = f_2 \\ u_x + v_y = g \end{cases}$$

[I.] MAC : Marker and Cell.



① in order to satisfy inf - sup condition

② for FDM to solve Stokes equation

* nothing new except the structure. all the rest are the same as Finite Difference Method. e.g. dealing with Laplacian operator and divergence operator.

[2.] DGS: Distributive Gauss-Seidel.

① DGS is a smoother used in the multigrid method for solving Stokes equation by MAC structure.

② Why not standard relaxation, i.e. G-S relaxation?

the discrete structure: $\begin{pmatrix} A & B^T \\ B & 0 \end{pmatrix} \begin{pmatrix} \vec{u}_h \\ p_h \end{pmatrix} = \begin{pmatrix} \vec{f}_h \\ 0 \end{pmatrix}$

this guy is not diagonally dominant

zero block hampers the relaxation.

$\boxed{\mathcal{L} x = b}$

idea of DGS:

our principle operator: $\mathcal{L} = \begin{pmatrix} A & B^T \\ B & 0 \end{pmatrix}$.

let $\mathcal{M} = \begin{pmatrix} I & B^T \\ 0 & -BB^T \end{pmatrix}$

s.t. $\mathcal{L}\mathcal{M} = \begin{pmatrix} A & AB^T - B^TBB^T \\ B & BB^T \end{pmatrix}$ ≈ 0 in the sense of $\begin{cases} \text{① small} \\ \text{② low rank.} \end{cases}$ ✓

can be verified that $\text{rank}(AB^T - B^TBB^T) \le n^2 - (n-1)^2$

comparing with the rank of n^2, $= O(n^2)$

$n^2 - (n-1)^2 = (n+n-1)(n-n+1) = 2n-1 = O(n)$ relatively small.

$$\tilde{L}M = \begin{pmatrix} A & 0 \\ B & BB^T \end{pmatrix} \approx LM$$

approximation.

$$\therefore L^{-1} = M(LM)^{-1} \approx M(\tilde{L}M)^{-1}$$

so, the iterative method (residual-correction)

$$x^{k+1} = x^k + \underbrace{\underbrace{M(\tilde{L}M)^{-1}}_{\text{estimate } L^{-1}} \underbrace{(b - Lx^k)}_{\text{residual}}}_{\text{correction}}$$

update form.

basically in Multigrid Method, we prefer solving residual equation. Here using DGS is the same.

i.e. we use $L^{-1} \approx M(\tilde{L}M)^{-1}$ so $L \approx (\tilde{L}M) \cdot M^{-1}$

so $\boxed{Le = r} \longrightarrow (\tilde{L}M) \cdot M^{-1}e = r$ $\quad \tilde{e} := M^{-1}e$

$$\tilde{L}M \cdot \tilde{e} = r \qquad A_p := BB^T$$

$$\boxed{\begin{pmatrix} A & 0 \\ B & A_p \end{pmatrix} \tilde{e} = r}$$

solve this,

got $\tilde{e}$, the desired $e = M\tilde{e}$

this is why called "distributive"

DGS details: $\hat{A} \sim A$, $\hat{A}_p \sim A_p = BB^T$.

① relax the momentum equation.

$$u^{k+\frac{1}{2}} = u^k + \underbrace{\underbrace{\hat{A}^{-1}}_{\text{estimate } A^{-1}} \underbrace{(f - Au^k - B^T p^k)}_{\text{residual}}}_{\text{correction}}$$

update. however, this is not $u^k + e_u$, this is $u^k + \tilde{e}_u$, that's why we denote it by $u^{k+\frac{1}{2}}$ instead of u^{k+1}. this is an intermedia term for distribution later.

② relax the mass equation.

$$e_p = \underbrace{\hat{A}_p^{-1}}_{\text{estimate } A_p^{-1}} \underbrace{(g - Bu^{k+\frac{1}{2}})}_{\text{residual}}$$

here we use new $u^{k+\frac{1}{2}}$ instead of u^k

③ distribute corrections to original variables.

$$\begin{cases} u^{k+1} = \underline{u^{k+\frac{1}{2}}} + \underline{B^T e_p} \\ p^{k+1} = p^k - \underline{BB^T e_p} \end{cases}$$

as $M\tilde{e} = e$. $M = \begin{pmatrix} I & B^T \\ 0 & -BB^T \end{pmatrix}$

unrelated to e_u so update directly.